Problem set 1; Solutions: Special relativity

1. **Some properties of vectors in Minkowski spacetime**

   (a) Let $\boldsymbol{u} \in \mathcal{M}$ a timelike vector and $\boldsymbol{v} \in \mathcal{M}$ such that $\boldsymbol{\eta}(\boldsymbol{u}, \boldsymbol{v}) = 0$. Then, by definition, we have:

   $$\begin{aligned} -\left(u^0\right)^2 + \vec{u} \cdot \vec{u} &< 0 \\ -u^0 v^0 + \vec{u} \cdot \vec{u} &= 0 \,, \end{aligned}$$

   which leads to:

   $$\begin{aligned} \left(u^0\right)^2 &> \vec{u} \cdot \vec{u} \\ (\vec{u} \cdot \vec{v})^2 &= \left(u^0\right)^2 \left(v^0\right)^0 \,. \end{aligned}$$

   The Cauchy-Schwartz inequality reads:

   $$(\vec{u} \cdot \vec{v})^2 \leq (\vec{u} \cdot \vec{u})\,(\vec{v} \cdot \vec{v}) \,.$$

   Thus:

   $$\left(u^0\right)^2 \left(v^0\right)^2 \leq (\vec{u} \cdot \vec{u})\,(\vec{v} \cdot \vec{v}) < \left(u^0\right)^2 (\vec{v} \cdot \vec{v}) \,.$$

   Since $u^0 \neq 0$, we get:

   $$\left(v^0\right)^2 < \vec{v} \cdot \vec{v} \,,$$

   which shows that $\boldsymbol{v}$ is spacelike.

   (b) Let us pick two future-directed timelike vectors $\boldsymbol{u}$ and $\boldsymbol{v}$:

   $$\begin{aligned} \boldsymbol{\eta}\,(\boldsymbol{u}, \boldsymbol{u}) < 0 \;\text{ with }\; u^0 > 0 \\ \boldsymbol{\eta}\,(\boldsymbol{v}, \boldsymbol{v}) < 0 \;\text{ with }\; v^0 > 0 \,, \end{aligned}$$

   so that $\boldsymbol{\eta}\,(\boldsymbol{u}, \boldsymbol{v}) < 0$. Then:

   $$\boldsymbol{\eta}\,(\boldsymbol{u} + \boldsymbol{v}, \boldsymbol{u} + \boldsymbol{v}) = \boldsymbol{\eta}\,(\boldsymbol{u}, \boldsymbol{u}) + \boldsymbol{\eta}\,(\boldsymbol{v}, \boldsymbol{v}) + 2\boldsymbol{\eta}\,(\boldsymbol{u}, \boldsymbol{v}) < 0 \,,$$

   which shows that $\boldsymbol{u} + \boldsymbol{v}$ is timelike. Moreover, $(u+v)^0 = u^0 + v^0 > 0$, so that $\boldsymbol{u} + \boldsymbol{v}$ is future-directed.

   (c) Let $\boldsymbol{d} \in \mathcal{M}$ be a spacelike vector. We write:

   $$\boldsymbol{d} = d^0 \boldsymbol{e}_{(\mathbf{0})} + \left|\vec{d}\right| \boldsymbol{n} \,,$$

   with

   $$\boldsymbol{n} = \frac{\vec{d}}{\left|\vec{d}\right|}$$

a unit spacelike vector. If $d^0 \neq 0$, using a spacetime diagram, we are led to define two lightlike vectors:

$$\begin{aligned} \boldsymbol{l} &= \boldsymbol{\eta}\left(\boldsymbol{d}, \boldsymbol{e}_{(\mathbf{0})}\right)\left[\boldsymbol{e}_{(\mathbf{0})} + \boldsymbol{n}\right] \\ \tilde{\boldsymbol{l}} &= \boldsymbol{\eta}\left(\boldsymbol{d}, \boldsymbol{e}_{(\mathbf{0})}\right)\left[\boldsymbol{e}_{(\mathbf{0})} - \boldsymbol{n}\right] \end{aligned}$$

which are such that $l^0 = \tilde{l}^0 = d^0$. Then, a bit of algebra shows that:

$$\boldsymbol{d} = \frac{d^0 + \left|\vec{d}\right|}{2\boldsymbol{\eta}\left(\boldsymbol{d}, \boldsymbol{e}_{(\mathbf{0})}\right)}\boldsymbol{l} - \frac{-d^0 + \left|\vec{d}\right|}{2\boldsymbol{\eta}\left(\boldsymbol{d}, \boldsymbol{e}_{(\mathbf{0})}\right)}\tilde{\boldsymbol{l}} \ .$$

Thus, defining:

$$\begin{aligned} \boldsymbol{k} &= \frac{d^0 + \left|\vec{d}\right|}{2\boldsymbol{\eta}\left(\boldsymbol{d}, \boldsymbol{e}_{(\mathbf{0})}\right)}\boldsymbol{l} \\ \tilde{\boldsymbol{k}} &= \frac{-d^0 + \left|\vec{d}\right|}{2\boldsymbol{\eta}\left(\boldsymbol{d}, \boldsymbol{e}_{(\mathbf{0})}\right)}\tilde{\boldsymbol{l}} \ , \end{aligned}$$

we have that $\boldsymbol{d}$ is the difference of two lightlike vectors:

$$\boldsymbol{d} = \boldsymbol{k} - \tilde{\boldsymbol{k}} \ .$$

And since:

$$\boldsymbol{\eta}\left(\boldsymbol{d}, \boldsymbol{d}\right) = -\left(d^0\right)^2 + \left|\vec{d}\right|^2 > 0 \ ,$$

we have:

$$\begin{aligned} \boldsymbol{\eta}\left(\boldsymbol{k}, \boldsymbol{e}_{(\mathbf{0})}\right) &= -\frac{d^0 + \left|\vec{d}\right|}{2} < 0 \\ \boldsymbol{\eta}\left(\tilde{\boldsymbol{k}}, \boldsymbol{e}_{(\mathbf{0})}\right) &= \frac{d^0 - \left|\vec{d}\right|}{2} < 0 \end{aligned}$$

which shows that both lightlike vectors are future directed.
If $d^0 = 0$, then one can simply define:

$$\begin{aligned} \boldsymbol{l} &= \boldsymbol{e}_{(\mathbf{0})} + \boldsymbol{n} \\ \tilde{\boldsymbol{l}} &= \boldsymbol{e}_{(\mathbf{0})} - \boldsymbol{n} \ , \end{aligned}$$

and clearly:

$$\boldsymbol{d} = \frac{\left|\vec{d}\right|}{2}\left(\boldsymbol{l} - \tilde{\boldsymbol{l}}\right) \ ,$$

which gives the expected result.

2. **Relative Lorentz factor**
Let us call F the inertial frame with coordinates $x^\mu = (t, x, y, z)$. The 4-velocity of O_2 wrt F is given by:

$$\boldsymbol{u_2} = \gamma_2\left[\frac{\partial}{\partial t} + \vec{v}_2\right] = u_2^\mu \boldsymbol{e}_{(\mu)} \ ,$$

where $\gamma_2 = 1/\sqrt{1-v_2^2}$ and $v_2 = |\vec{v}_2|$. The components of this 4-velocity in the inertial frame attached to O_1 are given by:

$$\tilde{u}^\mu = \Gamma^\mu{}_\nu\left(\vec{v}_1\right) u_2^\nu \ ,$$

where $\Gamma^\mu{}_\nu\left(\vec{v}_1\right)$ are the components of the Lorentz transformation from the inertial frame attached to O_1 into F. By definition, the Lorentz factor of O_2 wrt O_1 is given by:

$$\gamma_{12} = \tilde{u}_2^0 \ .$$

Thus:

$$\begin{aligned}
\gamma_{12} &= \tilde{u}_2^0 \\
&= \Lambda^0{}_0 u_2^0 + \Lambda^0{}_i u_2^i \\
&= \gamma_1\gamma_2 - \beta_1\gamma_1\gamma_2 \vec{d}\cdot\vec{v}_2 \\
&= \gamma_1\gamma_2\left[1 - \vec{v}_1\cdot\vec{v}_2\right] \ .
\end{aligned}$$

3. **Some orders of magnitude**
For $\beta \ll 1$, we get:

$$\begin{aligned}
\gamma &= 1 + \frac{1}{2}\beta^2 + O\left(\beta^4\right) \\
\beta\gamma &= \beta + O\left(\beta^3\right) .
\end{aligned}$$

Therefore, we have at the lowest order in β:

$$\begin{aligned}
x^0 &= \hat{x}^0 + \beta\hat{x}^1 \\
x^1 &= \hat{x}^1 + \beta\hat{x}^0 .
\end{aligned}$$

Now, re-establishing the c's, we get $x^0 = ct$, $\hat{x}^0 = c\hat{t}$ and $\beta = v/c$, so that:

$$\begin{aligned}
t &= \hat{t} + \frac{v}{c^2}\hat{x}^1 \\
x^1 &= \hat{x}^1 + v\hat{t} .
\end{aligned}$$

One sees that the term $\frac{v}{c^2}\hat{x}^1$ is of order v^2/c^2 with $c \gg v$, so we have, at the lowest order:

$$\begin{aligned}
t &= \hat{t} \\
x^1 &= \hat{x}^1 + v\hat{t} ,
\end{aligned}$$

which is exactly the expression for a Galilean transformation. The same applies for the law of addition of velocities: $\beta = \beta_1 + \beta_2$ at the lowest order.
The time dilation between two frames in relative uniform motion at the speed β is given by:

$$\Delta\hat{x}^0 = \gamma\Delta x^0 = \left[1 + \frac{1}{2}\beta^2\right]\Delta x^0 + O\left(\beta^4\right) .$$

Therefore, the time dilation $\frac{\hat{t}-t}{t} = \frac{\Delta t}{t}$ is given, to the lowest relativic order, by:

$$\frac{\Delta t}{t} \sim \frac{1}{2}\left(\frac{v}{c}\right)^2 .$$

Here are the different cases:

- **Usain Bolt**: $v \sim \frac{100\ \mathrm{m}}{9.58\ \mathrm{s}} \sim 10.44\ \mathrm{m.s^{-1}}$. Then:

$$\frac{\Delta t}{t} \sim 6.13 \times 10^{-16},$$

which means that in 10 s, he has accumulated a time delay of:

$$\Delta t \sim 10^{-15}\ \mathrm{s} = 10\ \mathrm{fs}.$$

- **Car**: $v \sim 100\ \mathrm{km.h^{-1}}$. Then:

$$\frac{\Delta t}{t} \sim 4.33 \times 10^{-15},$$

which means that in 1 h, it has accumulated a time delay of:

$$\Delta t \sim 10^{-11}\ \mathrm{s} = 10\ \mathrm{ps}.$$

- **Intercontinental flight**: $v \sim 900\ \mathrm{km.h^{-1}}$. Then:

$$\frac{\Delta t}{t} \sim 3.47 \times 10^{-13},$$

which means that in 10 h, it has accumulated a time delay of:

$$\Delta t \sim 10^{-8}\ \mathrm{s} = 10\ \mathrm{ns}.$$

- **Supersonic aircraft**: $v \sim 680\ \mathrm{m.s^{-1}}$ (Mach 2). Then:

$$\frac{\Delta t}{t} \sim 2.58 \times 10^{-12},$$

which means that in 10 h, it has accumulated a time delay of:

$$\Delta t \sim 10^{-1}\ \mu\ \mathrm{s} = 100\ \mathrm{ns}.$$

4. **Relativistic Doppler effect**
   Simply follow the reasoning here. Calculations are elementary.

5. **4-acceleration**
   Simple calculations. If you encountered problems, come with your work and show it to me.

6. **Variational principle and free motion**
   Idem.

7. **Electromagnetic field**
   Idem.

8. **Accelerated frames**
   It is important to notice that the orthonormal rest frame of the observer is an instantaneous frame: it varies from event to event along the observer's worldline. In classical mechanics this is called a moving frame: $\left\{\vec{e_\mu}\left(\tau\right)\right\}$. This explains why at every value of τ it is an inertial frame at that value of the proper time, around the observer, even though it is not a global inertial frame. There exists a Lorentz transformation between $\left\{\vec{e_\mu}\left(\tau\right)\right\}$ and $\left\{\vec{e_\mu}\left(\tau'\right)\right\}$ for any two values of τ and τ', by composition of the infinitesimal ones used to generate the frame, but these have nothing to do with the observer's proper motion.

9. **A beam of pions**
We call F the frame of the pion (i.e. the frame in which the pion is at rest), and $\hat{F}$ the frame of the laboratory.

(a) In the frame of the pion, we have: $\frac{\Delta x^0}{c} = 1.8 \times 10^{-8}$ s. In the frame of the lab, because of the time dilation, we get:

$$\Delta \hat{x}^0 = \gamma \Delta x^0.$$

For $\beta = 0.99$, $\gamma \sim 7.07$, and therefore:

$$\frac{\Delta \hat{x}^0}{c} = 1.27 \times 10^{-7} \text{ s}.$$

(b) We also have:

$$\Delta \hat{x}^1 = \gamma \Delta x^1 + c\beta\gamma \Delta x^0 = c\beta\gamma \Delta x^0,$$

because the pion is at rest in F, so that $\Delta x^1 = 0$. Hence:

$$\Delta \hat{x}^1 = c \times 0.99 \times 1.27 \times 10^{-7} \text{ s},$$

or:

$$\Delta \hat{x}^1 \sim 37.7 \text{ m}.$$

10. **Muon showers**
Let us call F the frame of the muon and $\hat{F}$ the frame of the Earth. Let us call x_0 the event: 'the muon is created', and x the event 'the muon reaches the surface of the Earth'. Let us also denote $\Delta x = x - x_0$ their difference vector.

(a) In a Galilean invariant theory, we would have: $\Delta x^0 = \Delta \hat{x}^0$. Then, by writing $\Delta t = \Delta x^0/c$ and the same for hatted quantities, we have: $\Delta \hat{t} = \Delta t$, so that we can apply the same formula in the two frame (and not only in the frame in which the pion is at rest), and the intensity at the surface of the Earth would be:

$$I\left(\Delta \hat{t}\right) = I_0 \exp\left(-\frac{\Delta \hat{t}}{\tau}\right).$$

Now, because the theory is Newtonian we also have simply that: $\hat{t} = \frac{L}{0.99 \times c}$. Therefore, $\Delta \hat{t} \sim 5 \times 10^{-5}$ s, and:

$$I\left(\Delta \hat{t}\right) \sim 10^{-10} I_0.$$

Hence, most of the muons have disappeared when the beam attains the surface of the Earth, making the detection of muons virtually impossible.

(b) Consider the problem in the frame of the muon. We construct a spacetime diagram; *cf* Fig. 1. The origins of the frames F and $\hat{F}$ are both taken at the event x_0. In particular, since the muon is at rest in F, its worldline is simple the x^0-axis. In the frame of the Earth, the surface of the Earth is the point y situated at $\hat{x}^1 = -L$; it is represented by the line (S). It moves towards the muon at the speed β (it is parallel to the $\hat{x}^0$-axis).

The event x is therefore the intersection of the x^0-axis with (S). The measurement of the distance between the top of the atmosphere and the surface in F must be done at $x^0 = \text{cst}$, so that it corresponds to the distance $[0z]$: there is a contraction of length:

$$Oz = \frac{Oy}{\gamma} = \frac{L}{\gamma}.$$

Now, the corresponding Δt in F, i.e. the time it takes for the surface of the Earth to reach the muon is read through $0x$. Since the equation for (S) is $x^0 = \frac{1}{\beta} x^1 + c\Delta t$, we find, at $x^0 = 0$ and $x^1 = 0z$:

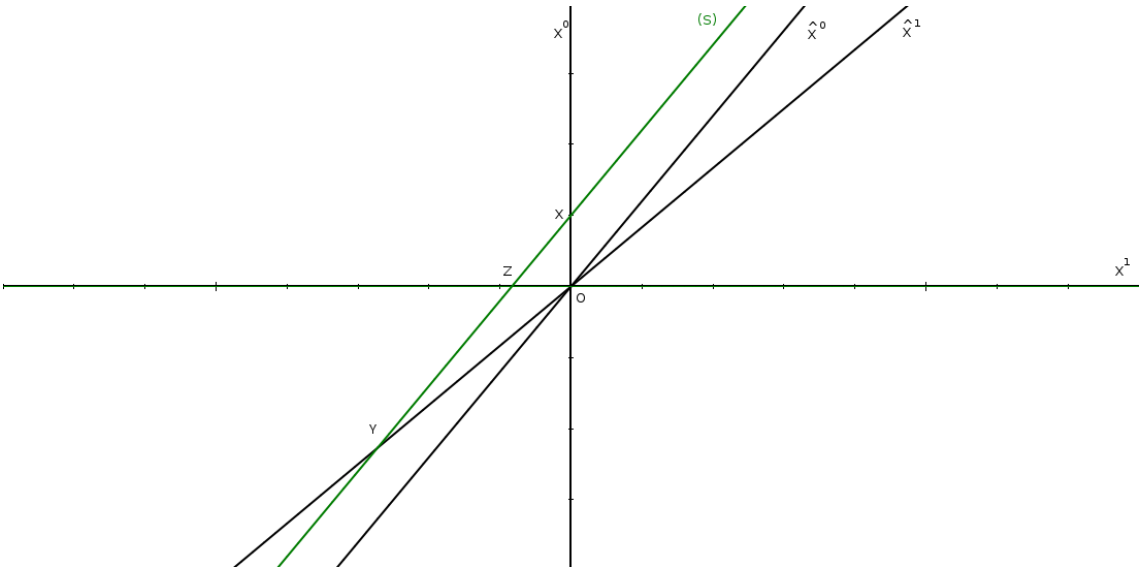


Figure 1: Spacetime diagram in the reference frame of the muon.

$$\Delta t = \frac{L}{c\beta\gamma}.$$

Since we are in the frame of the muon, we can apply the formula for the intensity, and the intensity at the surface is now:

$$I\left(\Delta t\right) = I_0 \exp\left(-\frac{L}{c\beta\gamma\tau}\right),$$

and therefore: $I\left(\Delta t\right) \sim 3.96 \times 10^{-2} I_0$. This is not so small and may even be detectable!

(c) Consider now the same problem in the frame of the Earth. Because the muon is at rest in F, we have that:

$$\Delta\hat{x}^0 = \Lambda^0{}_0 \Delta x^0 = \gamma \Delta x^0.$$

Moreover, in the frame of the Earth, the motion of the muon obeys: $\Delta\hat{x}^1 = L = \beta\Delta\hat{x}^0$ (in a spacetime diagram where the hatted coordinates corresponding to the frame of the Earth are now at a right angle, the worldline of the muon in a straight line parallel to the x^0-axis, i.e. a line with slope β^{-1}). Therefore:

$$\Delta x^0 = \frac{L}{\beta\gamma},$$

and the time lapse in the muon frame is $\Delta t = \frac{L}{c\beta\gamma}$, and we recover the previous result.

(d) We see that, as expected, the calculation of the observable, i.e. the intensity of the muon beam, does not depend on which frame is used to make the calculation. What appears as a time dilation in the Earth's frame appears as a length contraction in the muon frame, and both calculations lead to the same result.